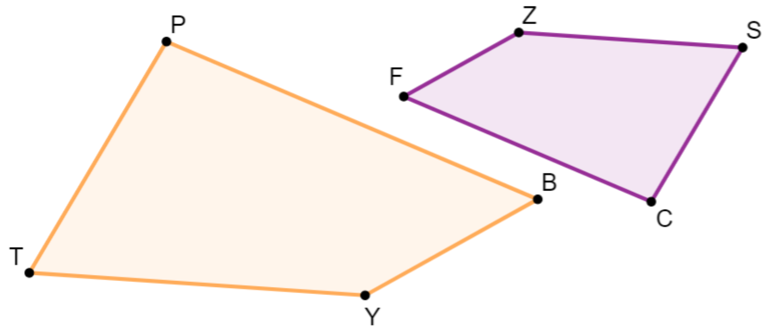
6.11.4 Wrap-Up: Error Analysis

1. Cheryl and Austin are finding the measure of side SC. They were given the following information about similar quadrilaterals $ZSCF$ and $YTPB$ that are shown. $\Delta ZSCF \sim \Delta YTPB, ZS = 15$, $SC = 5x - 8$, $PT = 3x + 6$, and $YT = 22.5$.



Cheryl's and Austin's equations are shown.

Cheryl's Work	Austin's Work
$15(5x - 8) = 22.5(3x + 6)$	$1.5(5x - 8) = 3x + 6$

a. Determine whose equation, if any, is set up incorrectly. Show your work or write an explanation.

b. What is the length of $\overline{SC}$?

Unit 6, Lesson 12: Solving Problems Using Similarity – Part 2

Benchmark

MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.

Learning Target

☐ I can solve real-world problems involving similarity in two-dimensional figures.

6.12.1 Warm-Up: Biomedical Career

Liz Williams and Kate Lomas co-founded the Australian-based biomedical technology company, Hemideina. In Kate's studies she found that the insect wētā has an amazing hearing system. Through her studies she invented an implant that can help those with hearing loss live a life without limits.

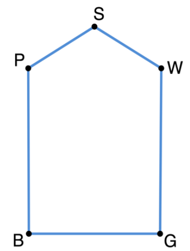
1. List at least one advantage that Kate may have had as an artist that helped her create her invention.

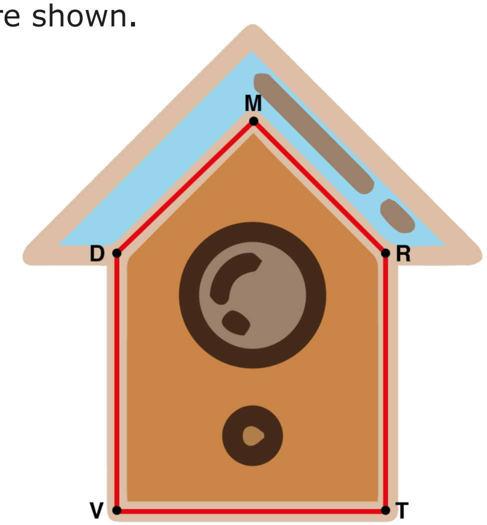
Both Liz and Kate express excitement knowing that that their skills in science can be used to solve real-world problems.

2. What is one real-world problem that you think creativity, math, and science could be used to solve?

6.12.2 Guided Instruction: Similar Figures in the Real World

1. Haley built a birdhouse by first drawing a blueprint. She then constructed the birdhouse using the dimensions from the blueprint. Her blueprint, $PSWGB$, and birdhouse, $DMRTV$, are shown.





To build the birdhouse, Haley multiplied each of the dimensions from the blueprint by 4 to get the dimensions of the face of the birdhouse. She used equations to represent the sides.

$PS = (y + 1.2)$ inches (in.), $PB = (3x - 1.5)$ in., $BG = (2w - 6.4)$ in., $VT = (4w - 5.6)$ in., $DV = (4x + 10)$ in., and $DM = (6.2y + 2.6)$ in.

a. What is the value of x?

b. What is the length of $\overline{DV}$? Include units.

c. What is the value of y?

d. What is the length of $\overline{PS}$? Include units.

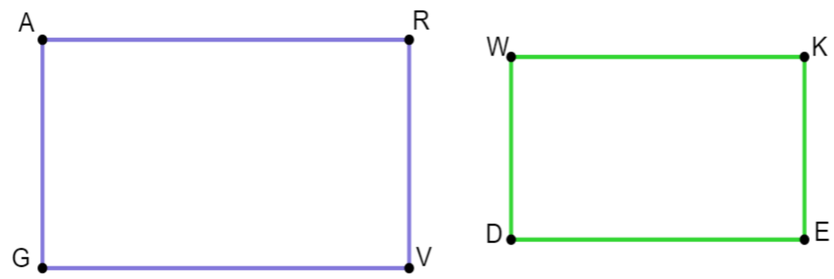
e. What is the value of w?

f. What is the length of $\overline{VT}$? Include units.

g. If $DV = RT$ and $DM = MR$, what is the perimeter, in inches, of the face of the birdhouse?

6.12.3 Guided Practice: Similar Figures in the Real World

1. Moving-R-Us creates boxes in the shape of a rectangular prism. The company uses a computer program to calculate similar size boxes by using equations. The diagram shows the rectangular bases, $ARVG$ and $WKED$, of two different size boxes, where $WD = 2$ inches (in.), $AR = (7w - 30)$ in., $RV = 2.5$ in., and $WK = (6w - 26)$ in.



a. What is the value of w?

b. What is the area of $ARVG$? Include units.

c. What is the perimeter of $WKED$? Include units.

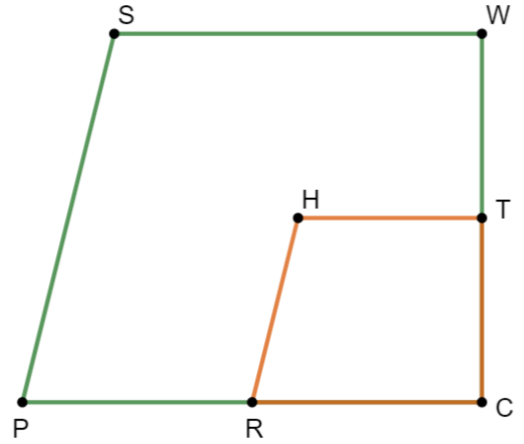
Moving-R-Us created a third similar rectangular box. The base, $MHST$, is shown, where $MH = 6.25$ in.

d. What is the length of $\overline{MT}$? Include units.

e. What is the perimeter, in inches, of $MHST$?

6.12.4 Try It: Solving in Real-World Context

1. Gloria, a landscape engineer, created a trapezoidal-shaped garden with a similar trapezoidal-shaped garden inside. A diagram of the gardens, $SWCP$ and $HTCR$, are shown, where $PC = (3y + 4)$ feet (ft), $RC = (y + 3)$ ft, $SP = 8.2$ ft, $SW = (7x - 13)$ ft, $HT = (2x - 2)$ ft, and $HR = 4.1$ ft.



a. What is the value of x?

b. What is the length of side SW? Include units.

c. What is the value of y?

d. What is the length of side RC? Include units.

e. If $SW = WC$, what is the perimeter, in feet, of the larger garden?

After Gloria created the gardens, she decided to check the angle measures of each corner. She had the following information written down: $m\angle SWC = (12z + 6)°$, $m\angle HTC = (7z + 41)°$, $m\angle TCR = 90°$, $m\angle SPC = (5w + 11)°$, and $m\angle HRC = (4w + 24)°$.

f. What is the value of z?

g. What is the value of w?

h. What is the measure of $\angle RHT$?

6.12.5 Wrap-Up: Cool Down

1. Your friend is absent today and will need the notes from class. Write a quick note to let them know what they missed in today's lesson.